



Miniature IoT WATER TDS And LEVEL MONITOR Cum CONTROLLER

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For setups that rely on stored water, clear awareness of tank level and water quality is essential. Traditional level monitors are often bulky offline devices with only simple LED indicators, while many newer smart models can be expensive and still ignore key quality parameters such as total dissolved solids (TDS) and turbidity. As environmental contamination increases, continuous monitoring of both water level and purity has become necessary.

This system provides a cost-effective approach with remote pump control over Wi-Fi, data logging, and the ability to operate in mesh configurations across multiple tanks or monitoring points. It measures water level, TDS, and turbidity in real time, issues alerts when thresholds are crossed, and stores the readings for later review, offering significant savings compared to many commercial options.

Compact and easy to install, the system uses the IndusBoard Coin V2 as the central controller due to its small size and integrated analogue-to-digital conversion. Alerts are generated when the water becomes cloudy or shows increased dissolved solids, and all data is stored locally. Fig. 1 shows the prototype and dashboard, and the Bill of Materials table lists the required components.

Circuit and working

Fig. 2 shows the circuit diagram of the miniature IoT water TDS and level monitor cum controller system. The core elements include the IndusBoard Coin V2, a TDS sensor, a turbidity sensor, stainless

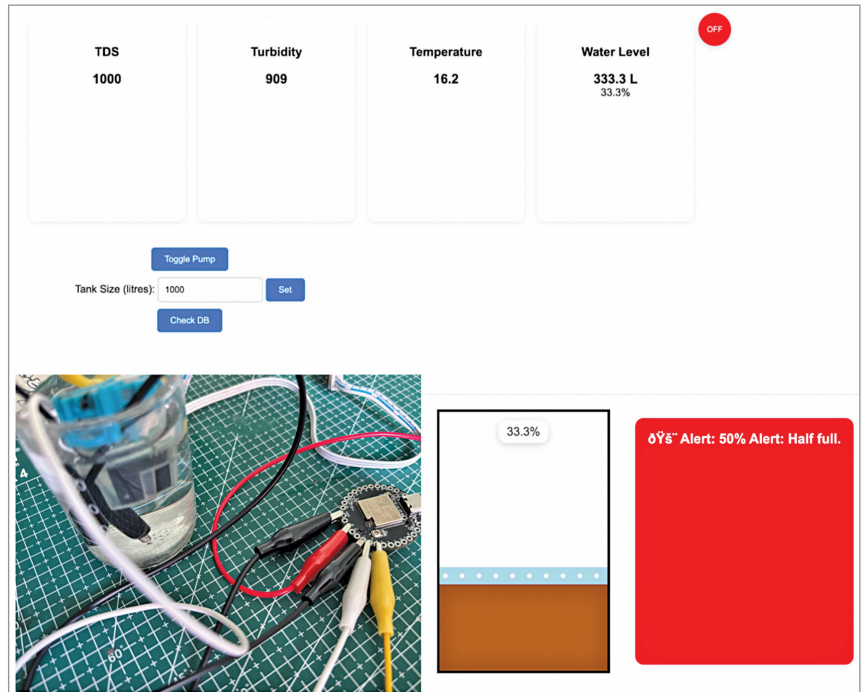


Fig. 1: Author's prototype and web dashboard

steel probes for water level sensing, and supporting components.

The water level is detected by measuring conductivity and capacitance across stainless steel probes placed at different heights in the tank. For tanks with uniform shapes, a time-of-flight laser distance sensor may be used for calculating volume more precisely.

Here, we have used I/O pins 3, 4, and 5 for water level detection. If you need finer level detection (e.g., 10%, 20%, 30% of tank fill), you can utilise additional free GPIO pins on the Coin board. The IndusBoard Coin provides over 30 I/O pins for

BILL OF MATERIALS

Components	Quantity	Description
IndusBoard Coin V2	1	ESP32-S2 brain (Wi-Fi, ADC, GPIO)
TDS sensor	1	Analogue probe for TDS (0-1000ppm)
Turbidity sensor	1	IR optical module (0-1000NTU)
Stainless probes	3	For level sensing (conductive strips)
5V single relay	1	Controls pump/valve (pin 33)
USB-C cable	1	Power/programming
5V power adaptor	1	5V power supply

flexibility. Pin 5 is used for the TDS and turbidity sensors, while pin 33 is connected to a relay to control the water pump or valves. Additional relays can also be added using the remaining free I/O pins.

The board and sensors operate from the 3V output, and the relay is powered via the VIN line. Power is supplied using a standard 5V adaptor through the USB port.